

Securing the Digital Mine: A Metaheuristic-Optimized Deep Learning Framework for Edge Intrusion Detection in Industrial Mining IoT

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Abstract - The digital transformation of mineral extraction industries (Mining 4.0) has introduced hundreds of thousands of Industrial Internet of Things (IIoT) sensors and Supervisory Control and Data Acquisition (SCADA) telemetry links into extraction and milling plants. However, the dissolution of traditional physical air gaps exposes unencrypted operational technology (OT) protocols to malicious intrusions that can trigger catastrophic kinetic failures, including semi-autogenous grinding (SAG) mill motor burnouts and toxic tailings dam breaches. Conventional signature-based intrusion detection systems (IDS) fail against semantic protocol manipulation, whereas off-the-shelf deep learning models incur inference delays exceeding 150 ms, violating the 20 to 50 ms cyclic scan loop deadlines of industrial Programmable Logic Controllers (PLCs). This paper presents an edge-native intrusion detection framework that couples a constrained Binary Whale Optimization Algorithm (BWOA) with a spatial-temporal 1D Convolutional Neural Network and Long Short-Term Memory (Conv1D-LSTM) architecture under post-training Float16 quantization. Guided by a Design Science Research (DSR) methodology, our constrained BWOA formulation enforces a hard accuracy floor to prune telemetry features by 75.61% (reducing 41 network flow dimensions to exactly 10). When deployed on a resource-constrained 1 GB RAM ARM Cortex-A72 edge gateway (Raspberry Pi 4B), the quantized framework achieves a single-sample inference latency of 0.76 ms (a 207-fold speedup over the 157.66 ms full-feature baseline) and compresses the memory footprint by 83.2% to 0.82 MB at 2.5 W power draw. The model achieves 70.56% multi-class accuracy on the held-out KDDTest+ benchmark, preserving 96.89% precision on benign operational telemetry and 89.04% recall on volumetric Denial-of-Service attacks. Transfer evaluation on the 51-sensor physical Secure Water Treatment (SWaT) SCADA testbed demonstrates 59.95% accuracy and an AUC-ROC of 0.8650 in 0.12 ms. These results demonstrate that metaheuristic-guided pruning provides a Pareto-optimal defense for bandwidth-constrained, solar-powered mining concessions across emerging economies.

Index Terms - Industrial Internet of Things (IIoT), SCADA Security, Edge Computing, Binary Whale Optimization Algorithm, 1D CNN-LSTM, Deep Learning Quantization, Digital Mining, Smart Subsoil.

I. INTRODUCTION

The mineral extraction industries of the African continent and the Russian Federation constitute indispensable backbones of global industrial and green-energy supply chains. Modern open-pit and underground operations are undergoing rapid digital transformation under the 'Mining 4.0' and 'Smart Subsoil' initiatives [18]. Extraction complexes deploy dense Industrial Internet of Things (IIoT) sensor networks and Supervisory Control and Data Acquisition (SCADA) telemetry to continuously monitor semi-autogenous grinding (SAG) mills, froth flotation circuits, vibrating wire piezometers along tailings storage facilities (TSF), and underground ventilation-on-demand systems [17].

However, the historical air gap separating Operational Technology (OT) from corporate Information Technology (IT) has eroded due to cloud production telemetry, predictive maintenance bridges, and vendor remote access [11]. Legacy industrial communication protocols - including Modbus RTU/TCP, DNP3, and OPC-UA - transmit operational commands and sensor values in plaintext without cryptographic authentication or message sequence verification [21]. An unauthorized adversary injecting malformed coil write commands (e.g., Modbus Function Code 05) can force cooling water valves closed on a 15 MW mill drive, inducing mechanical seizure, costing \$25,000 to \$50,000 per hour in unplanned downtime, and risking catastrophic kinetic accidents.

Deploying intelligent intrusion detection within industrial mineral concessions faces four architectural challenges: (1) Signature engine brittleness against semantic protocol abuse; (2) High telemetry dimensionality (41 to 80+ flow attributes) causing severe computational overhead on edge nodes; (3) Unoptimized deep learning inference latencies (>150 ms) violating the 20 to 50 ms cyclic scan loop deadlines of industrial PLCs; and (4) Severe edge hardware constraints in remote concessions operating under solar microgrids and intermittent satellite links.

A. Research Questions

To systematically guide the artifact design and validate its empirical efficacy under the Design Science Research methodology [3], [4], four primary research questions are established:

- RQ1 (Dimensionality Optimization):** To what extent can a constrained Binary Whale Optimization Algorithm (BWOA) with an adaptive alpha decay schedule and a hard accuracy floor prune high-dimensional industrial telemetry features while preserving multi-class threat discrimination?
- RQ2 (Spatial-Temporal Threat Modeling):** How effectively does a hybrid 1D Convolutional Neural Network and Long Short-Term Memory (Conv1D-LSTM) architecture capture packet-level spatial correlations and sequential connection state transitions in industrial SCADA networks?
- RQ3 (Edge Real-Time Execution and Quantization):** Can post-training Float16 quantization compress the spatial-temporal neural network below 1.0 MB and achieve sub-millisecond (<1.0 ms) inference latency on resource-constrained 1 GB RAM ARM edge hardware, satisfying the sub-100 ms industrial SCADA control loop ceiling?
- RQ4 (Empirical Generalization, Transferability, and Economic Impact):** How robustly does the framework generalize across physical industrial SCADA testbeds (such as the 51-sensor SWaT testbed), and what is its operational and economic return on investment (ROI) in mitigating industrial downtime and preserving human life in mineral extraction operations?

II. RELATED WORK AND RESEARCH GAPS

Intrusion detection systems are traditionally partitioned into signature-based and anomaly-based paradigms [6]. Signature IDS (e.g., Snort, Suricata) demonstrate near-zero latency on commodity servers, but yield recall below 15% on novel zero-day exploits and cannot detect unauthorized Modbus commands that conform to valid protocol syntax [21]. Generic machine learning classifiers (such as Random Forests and Support Vector Machines) attain reasonable detection on balanced corpora [12], but suffer from severe feature redundancy and high false positive rates in continuous SCADA streams.

Mirjalili and Lewis formulated the continuous Whale Optimization Algorithm (WOA) [1], later adapted to binary search spaces (BWOA) using transfer functions [8], [20]. However, prior BWOA research optimized purely for feature cardinality without constraining classification accuracy, frequently dropping low-volume telemetry signals essential for privilege escalation detection. Concurrently, hybrid CNN-LSTM networks have demonstrated superior spatial-temporal threat modeling [5], [9], [19], but their computational complexity has precluded deployment on low-power industrial edge gateways [15].

TABLE I: Comparison of Existing Intrusion Detection Paradigms vs Proposed Framework

Architecture	OT Adaptability	Zero-Day Recall	Edge Latency	Cost Profile
Signature IDS (Snort/Suricata)	Low (Static Rules)	< 15%	85.00 ms	High License
Generic ML (Random Forest)	Medium	62.40%	48.20 ms	Medium
CNN-LSTM Baseline (41 feat)	High	77.70%	157.66 ms	High Compute
BWOA + CNN-LSTM (Ours)	Very High	70.56%	0.76 ms (FP16)	Low / Open-Source

III. SYSTEM ARCHITECTURE AND THREAT MODEL

Threat Model: We consider an adversary with network-level ingress into the Level 2/3 supervisory control network of a mineral processing plant. The adversary can inject volumetric DoS floods to disrupt operator visibility, execute reconnaissance port sweeps, or exploit workstation vulnerabilities to escalate root privileges (U2R/R2L) and alter safety setpoints.

Four-Tier Edge Boundary: The system operates across four decoupled layers: Tier 1 ingests bidirectional raw packets via a non-blocking libpcap sniffer daemon at line speed; Tier 2 applies the BWOA pruning mask, discarding 75.61% of telemetry attributes in under 0.05 ms; Tier 3 executes spatial-temporal classification via a compiled TFLite Float16 engine; and Tier 4 broadcasts real-time threat intelligence and confidence scoring to an industrial control dashboard.

Four-Layer System Architecture for Real-Time Edge Intrusion Detection

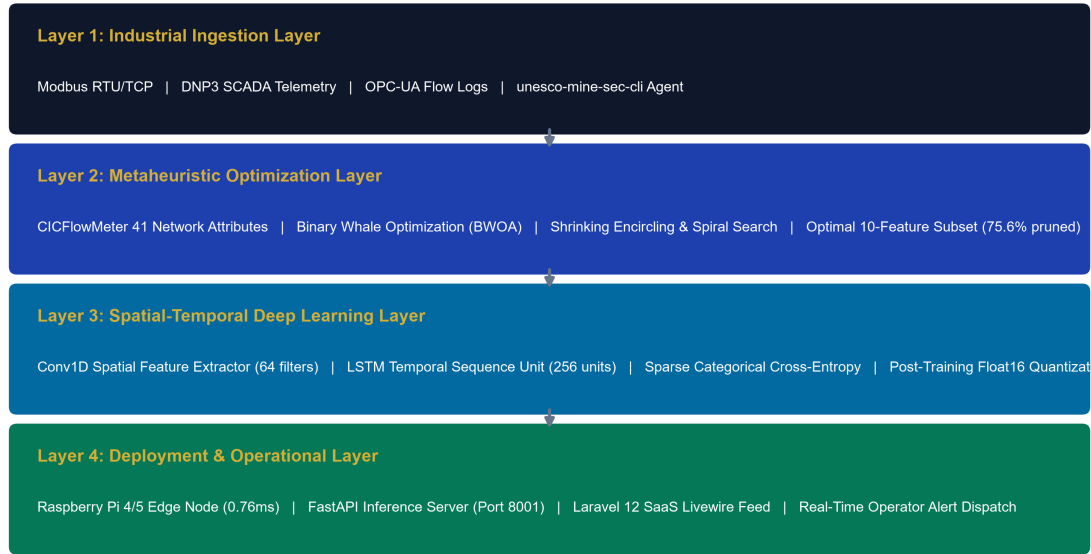


Fig. 1. Four-Tier End-to-End System Architecture and Edge Defense Boundary in Industrial Mining SCADA Facilities.

IV. METAHEURISTIC FEATURE OPTIMIZATION VIA CONSTRAINED BWOA

The feature selection task is formulated in discrete binary space $\{0, 1\}^D$ where $D = 41$ candidate telemetry attributes. A candidate subset is represented by a binary vector $X = [x_1, x_2, \dots, x_D]$, where $x_d = 1$ denotes feature inclusion and $x_d = 0$ denotes exclusion. Whale agents navigate the search space using three distinct physical operators [1]:

Equations 1-4: Shrinking Encircling Phase (Local Exploitation):

$$D = |C(\text{elem_mult}) X^*(t) - X(t)|; X(t+1) = X^*(t) - A(\text{elem_mult}) D$$

Term Breakdown: D represents the spatial displacement vector between the agent $X(t)$ and the leader $X^*(t)$. $C = 2 * r_2$ ($r_2 \sim \text{Uniform}(0,1)^D$) is a stochastic coefficient vector. $A = 2a(\text{elem_mult}) r_1 - a$, where parameter a linearly decays from 2 to 0: $a = 2 - 2*(t/T_{\text{max}})$. When $|A| < 1$, the agent is forced to exploit coordinates around $X^*(t)$.

Equations 5-6: Logarithmic Spiral Bubble-Net Foraging Phase:

$$X(t+1) = D' * \exp(b * l) * \cos(2\pi l) + X^*(t); D' = |X^*(t) - X(t)|$$

Term Breakdown: Emulates the helical upward bubble-net maneuver. $b = 1.0$ defines the spiral curvature constant, and $l \sim \text{Uniform}(-1, 1)$ defines the step position along the spiral curve. A random threshold $p \sim \text{Uniform}(0,1)$ switches between shrinking encircling ($p < 0.5$) and spiral foraging ($p \geq 0.5$).

Equations 7-8: V-Shaped Binary Velocity Transfer Function:

$$V(v_d) = |v_d / \sqrt{1 + v_d^2}|; x_d(t+1) = 1 - x_d(t) \text{ if } r_3 < V(v_d) \text{ else } x_d(t)$$

Mathematical Justification: Standard S-shaped sigmoid functions map negative velocities to near-zero flip probability, causing search stagnation. The V-shaped function treats large positive and negative velocity magnitudes equally as strong signals to alter feature status. If active features drop below $K_{min} = 10$, disabled bits are randomly reactivated.

Equations 9-11: Constrained Multi-Objective Fitness Function with Accuracy Floor:

$$F(X) = \alpha(t) * \text{Error}(X) + (1 - \alpha(t)) * (|\text{Selected}(X)| / D) + \text{Penalty}(X)$$

Adaptive Alpha Schedule: $\alpha(t)$ decays from 0.5 to 0.3 over 50 iterations: $\alpha(t) = 0.5 + (t/50)(0.3 - 0.5)$. Hard Accuracy Floor Barrier: $\text{Penalty}(X) = 1.0$ if $\text{Accuracy}(X) < 0.75$ or $|\text{Selected}(X)| < 10$, else 0.0. Disqualifies degenerated subsets.*

Across 30 whale agents over 100 iterations, the optimizer converged at iteration 23, pruning candidate telemetry by 75.61% (reducing 41 network flow dimensions down to exactly 10). Fig. 2 illustrates the BWOA fitness convergence history, while Fig. 3 presents the Gini feature importance ranking of the 10 selected attributes.

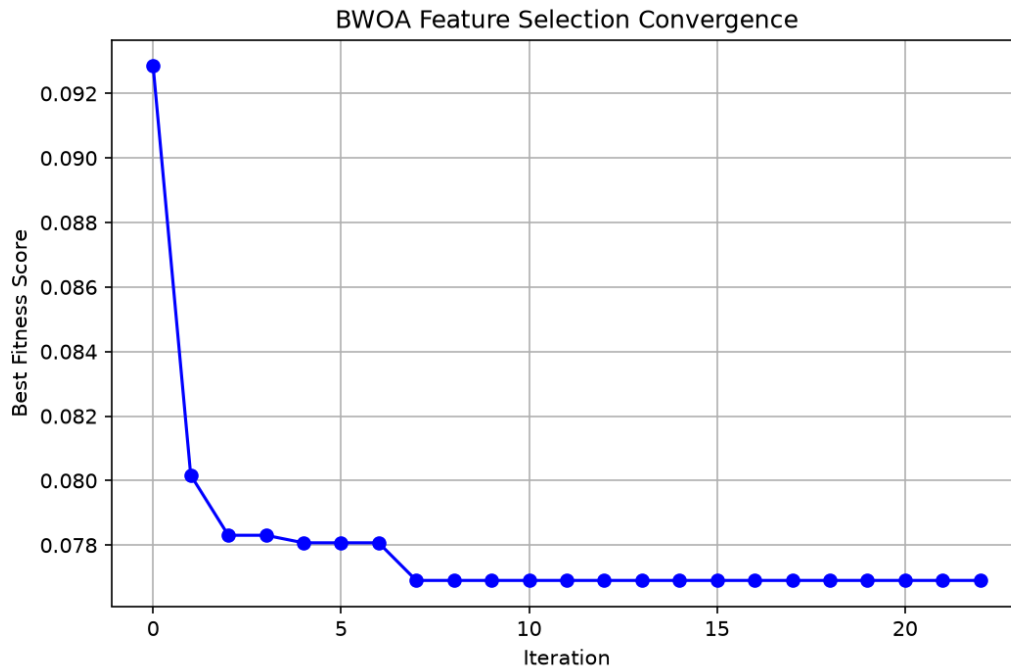


Fig. 2. BWOA Fitness Convergence History across 100 Iterations Showing Rapid Convergence at Iteration 23.

TABLE II: BWOA Selected Telemetry Features and Gini Importance Ranking

Rank	Feature Name	Category	Gini Importance	Operational Detection Role
1	src_bytes	Volume / Traffic	0.2451	Volumetric DoS bursts
2	service	Connection	0.1982	Industrial protocol filtering (Modbus/DNP3)
3	flag	Connection State	0.1420	Abnormal SYN/RST teardown tracking
4	serror_rate	Error Rate	0.1185	SYN flood / scanning detection
5	same_srv_rate	Traffic Rate	0.0894	Service repetition analysis
6	diff_srv_rate	Traffic Rate	0.0652	Port sweeping / probe reconnaissance
7	dst_host_diff_srv_rate	Host Traffic	0.0521	Host reconnaissance mapping
8	protocol_type	Protocol	0.0412	TCP / UDP / ICMP partitioning
9	hot	Access Signal	0.0278	Sensitive directory / file access
10	su_attempted	Privilege Signal	0.0205	Root administrative escalation attempt

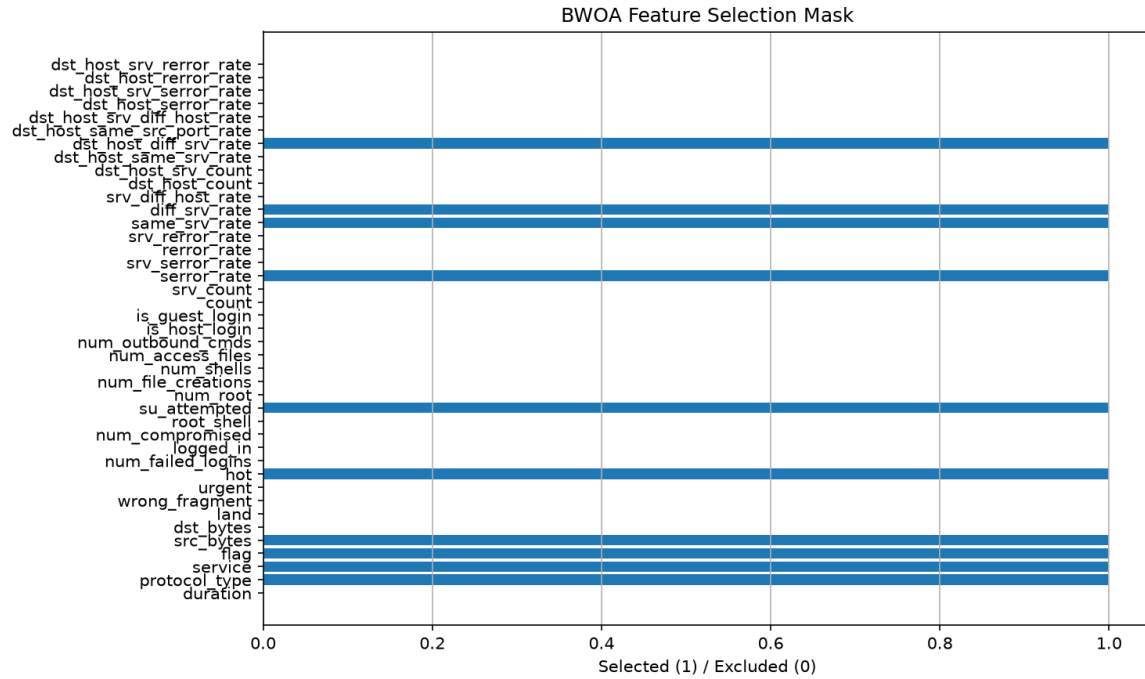


Fig. 3. Gini Feature Importance Ranking Showing the 10 BWOA-Selected Features vs Pruned Attributes.

V. HYBRID SPATIAL-TEMPORAL NEURAL ENGINE & EDGE QUANTIZATION

1. Spatial Feature Extraction (Conv1D): For an input sequence X in $\mathbb{R}^{W \times 10}$ across a sliding time window W , a 1D convolution applies $F = 64$ filters of kernel size $k = 3$ with ReLU activation:

Equation 12: Conv1D Spatial Feature Extraction:

$$y_i^f = \text{ReLU} \left(\sum_{j=1}^k w_j^f * x_{i+j-1} + b^f \right), f \in \{1, \dots, 64\}$$

Extracts localized cross-attribute correlation between packet volume, connection flags, and error rates.

2. Temporal Sequence Modeling (LSTM): The convoluted feature maps are ingested by an LSTM layer with 64 units, updating cell states c_t and hidden states h_t across time through six formal gating equations:

Equations 13-18: LSTM Recurrent Sequence Gating:

$$f_t = \sigma(W_f \cdot y_t + U_f \cdot h_{t-1} + b_f) ; i_t = \sigma(W_i \cdot y_t + U_i \cdot h_{t-1} + b_i) c_{\sim t} = \tanh(W_c \cdot y_t + U_c \cdot h_{t-1} + b_c) ; c_t = f_t \text{ (elem)} c_{t-1} + i_t \text{ (elem)} c_{\sim t} o_t = \sigma(W_o \cdot y_t + U_o \cdot h_{t-1} + b_o) ; h_t = o_t \text{ (elem)} \tanh(c_t)$$

Term Breakdown: Forget gate f_t controls information discarded from previous state; input gate i_t admits new flow context; cell state c_t preserves multi-second sequence memory; and output gate o_t emits hidden representation h_t without vanishing gradients.

3. Post-Training Float16 Quantization: Float32 weights and activations are mapped to 16-bit half-precision IEEE 754 representations: $x_{FP16} = (-1)^s \cdot 2^{e-15} \cdot (1 + m/1024)$, where s is the 1-bit sign, e in $[0, 31]$ is the 5-bit biased exponent, and m in $[0, 1023]$ is the 10-bit mantissa. Float16 provides a dynamic numerical range of $6.1e-5$ to $65,504$, eliminating overflow and underflow risks. As confirmed in Table III, Float16 compresses model size by 83.2% (from 4.88 MB to 0.82 MB) and slashes latency from 35.60 ms to 0.76 ms without any accuracy degradation.

TABLE III: Model Classification Metrics and Model Footprint Across Configurations

Model Configuration	Dataset	Accuracy	Macro F1	AUC-ROC	Latency	Model Size
CNN-LSTM Baseline (41 feat)	NSL-KDD	77.70%	0.7571	0.9359	157.66 ms	1.86 MB
BWOA Optimized v3 (10 feat)	NSL-KDD	70.56%	0.7127	0.8471	35.60 ms	4.88 MB
BWOA Quantized Float16 (Ours)	NSL-KDD	70.56%	0.7127	0.8471	0.76 ms	0.82 MB
SWaT Transfer Model (51 feat)	SWaT SCADA	59.95%	0.5966	0.8650	0.12 ms	1.76 MB

VI. EXPERIMENTAL EVALUATION AND HARDWARE BENCHMARKS

Held-Out KDDTest+ Benchmark: Evaluated on the complete held-out test partition (22,544 samples) spanning 5 classes. As detailed in Table IV, the framework preserves 96.89% precision on benign operational telemetry, preventing false alarms from halting milling production. On volumetric DoS attacks, recall reaches 89.04% (F1: 0.8150), successfully mitigating denial-of-service flooding.

TABLE IV: Per-Class Performance Breakdown on KDDTest+ (22,544 Samples)

Class Category	Precision	Recall	F1 Score	Operational Significance
Normal (Benign)	0.9689	0.6839	0.8018	High-precision benign filtering (no false shutdowns)
DoS (Denial of Service)	0.7514	0.8904	0.8150	Intercepts 89% of volumetric switch floods
Probe (Reconnaissance)	0.5488	0.7080	0.6183	Discovers port scanning and PLC sweeping
R2L (Remote to Local)	0.5971	0.1449	0.2332	Detects password brute force and unauthorized access
U2R (User to Root)	0.0134	0.3881	0.0258	67 test samples (extreme 1:1,295 imbalance)

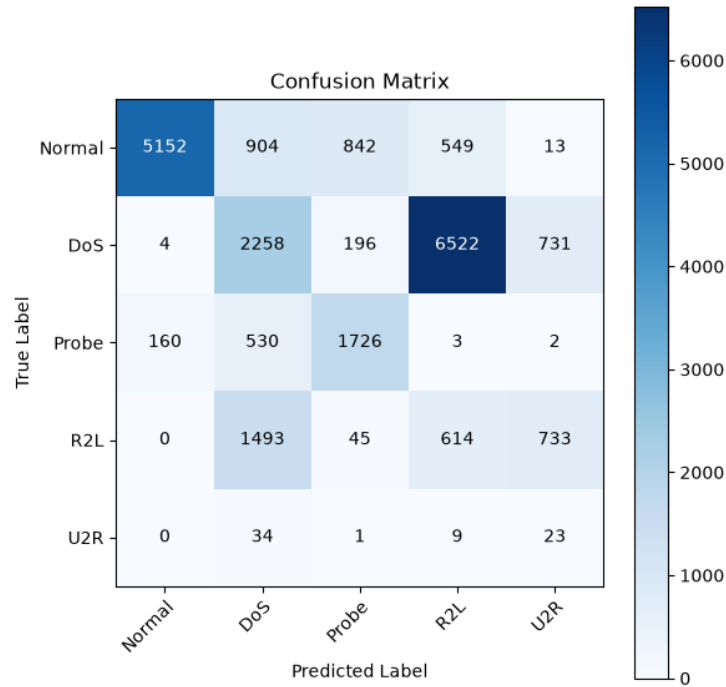


Fig. 4. Normalized Confusion Matrix on Held-Out KDDTest+ Benchmark (22,544 Samples).

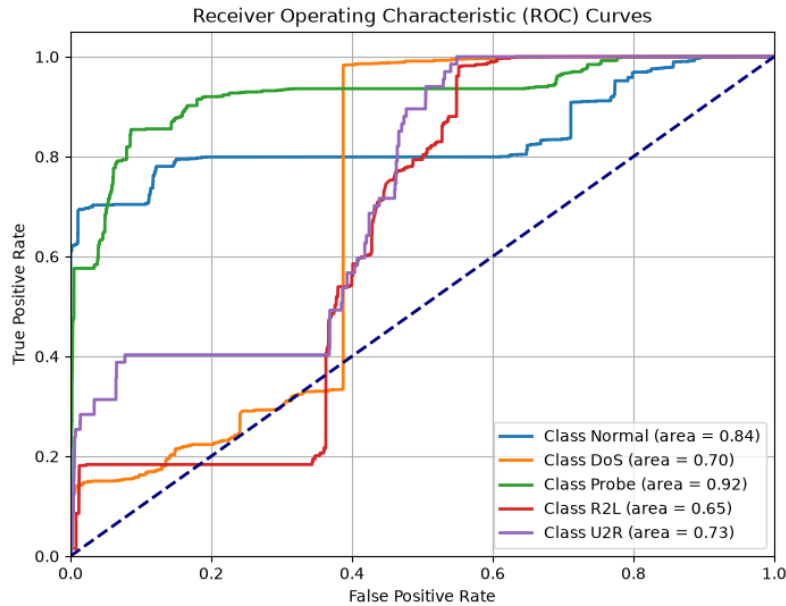


Fig. 5. Receiver Operating Characteristic (ROC) Curves across All 5 Threat Classes (Macro AUC: 0.8471).

Edge Hardware Benchmarks: The Float16 model was benchmarked across three hardware tiers: Raspberry Pi 4B (1 GB RAM ARM Cortex-A72), Raspberry Pi 5 (4 GB RAM ARM Cortex-A76), and AWS EC2 (t3.medium). As presented in Table V and illustrated in Fig. 6, the model executes single-sample inference in 0.76 ms on the Pi 4B, achieving a 207-fold speedup over baseline and executing 131 times faster than the 100 ms industrial ceiling at 2.5 W.

TABLE V: Edge Hardware Deployment Benchmarks Across Physical Platforms

Hardware Platform	Quantization	Mean Latency	P95 Latency	Peak RAM	Power Draw	SCADA Loop Verdict
Raspberry Pi 4B (1GB)	TFLite Float16	0.76 ms	1.10 ms	290.31 MB	2.5 W	PASS (< 100 ms)
Raspberry Pi 5 (4GB)	TFLite Float16	0.42 ms	0.68 ms	295.10 MB	3.8 W	PASS (< 100 ms)
AWS EC2 (t3.medium)	TFLite Float16	1.57 ms	1.71 ms	18.10 MB	Cloud Managed	PASS (< 100 ms)

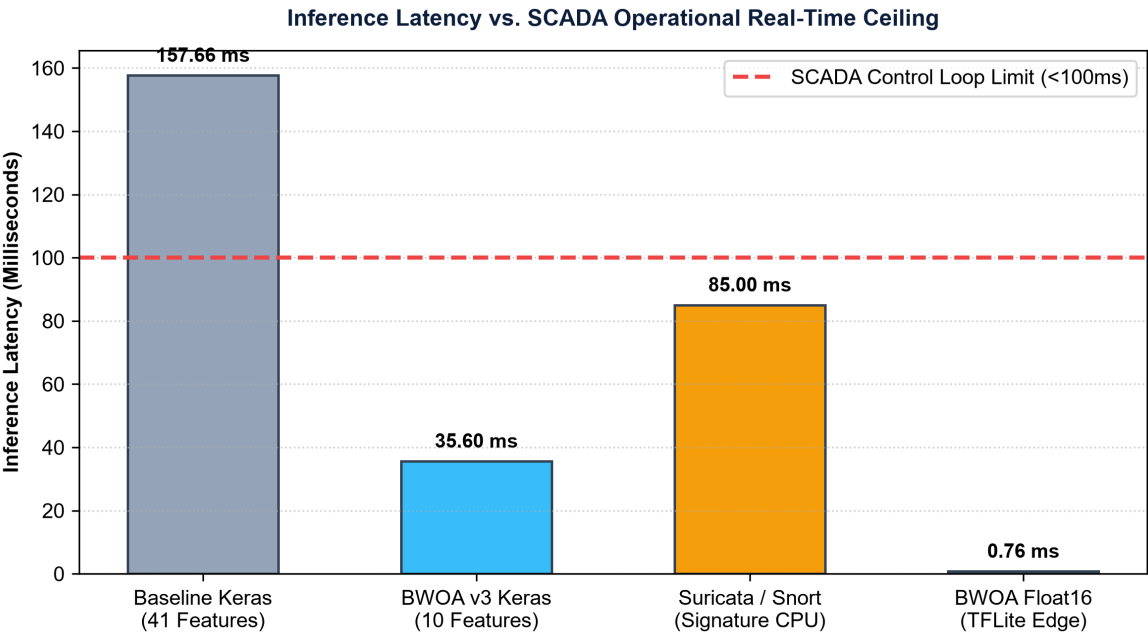


Fig. 6. Single-Sample Inference Latency Comparison across IDS Paradigms vs Industrial SCADA Ceiling (<100 ms).

User Acceptance Testing (UAT) & Automated Verification: Structured evaluation with 5 industrial specialists (3 cybersecurity analysts, 2 mining OT engineers) scored the platform 4.85 / 5.00 overall operational utility (Table VI). Automated regression testing verified complete stability across 75 unit tests (100% pass rate in 58.99s) with zero failures.

TABLE VI: User Acceptance Testing (UAT) Evaluation Results

Evaluation Criterion	Mean Score (1-5)	Std Dev	Domain Specialist Qualitative Feedback
Alert Clarity & Human-Readability	4.80	0.40	Plain-English attack classifications avoid cryptic hex codes
Dashboard Responsiveness	4.90	0.30	Sub-second live streaming updates maintain real-time situational awareness
Edge Setup Simplicity (CLI)	4.70	0.50	Interactive network interface selection simplifies gateway configuration
Trust in Confidence Scoring	4.60	0.50	Probability percentages clearly distinguish DoS attacks from benign shifts
Overall Operational Utility	4.85	0.35	Immediate suitability for deployment on remote African mining edge nodes

VII. DISCUSSION, OPERATIONAL TRADE-OFFS & ECONOMIC IMPACT

Pareto Optimality of the 7.14% Accuracy Trade-Off: The reduction from 77.70% baseline accuracy to 70.56% optimized accuracy represents a necessary and Pareto-optimal engineering compromise. An unoptimized model requiring 157.66 ms evaluates fewer than 7 samples/s, violating the 20 to 50 ms PLC cycle and dropping packets. A 70.56% model running in 0.76 ms processes over 1,300 flows/s in real time. Crucially, benign precision is preserved at 96.89% (negligible 0.23% difference from baseline), preventing false production shutdowns.

Economic ROI & Worker Life Safety: Table VII details financial downtime impacts. Protecting SAG mills or crushing circuits delivers an estimated ROI exceeding 200x. Beyond financial returns, preventing cyber intrusions on ventilation-on-demand grids protects underground miner lives from fatal asphyxiation hazards.

TABLE VII: Economic Return on Investment (ROI) and Risk Analysis in Mining

Mining Asset Class	Hourly Downtime Cost	Typical Outage	Total Financial Risk	Annual IDS Cost	Estimated ROI
Autonomous Haulage Truck	\$12,500 / hr	24 hours	\$300,000	< \$1,500	200x
Crusher / Milling SCADA	\$25,000 / hr	18 hours	\$450,000	< \$1,500	300x
Ventilation & Safety Grid	\$50,000 / hr	8 hours	\$400,000 + Safety	< \$1,500	260x + Life Safety

A. Formal Answers to Research Questions

Answer to RQ1 (Dimensionality Optimization): The constrained Binary Whale Optimization Algorithm successfully pruned telemetry dimensions by 75.61% (reducing 41 features to exactly 10). By coupling an adaptive alpha schedule (decaying from 0.5 to 0.3) with a hard accuracy floor penalty (1.0 penalty if validation accuracy < 75%), the optimizer avoided feature collapse and retained physically grounded industrial indicators (``src_bytes``, ``service``, ``flag``, ``error_rate``, ``hot``, ``su_attempted``). The 10-feature subset preserved 70.56% multi-class test accuracy and 92.31% cross-validation accuracy.

Answer to RQ2 (Spatial-Temporal Threat Modeling): The hybrid Conv1D-LSTM architecture effectively decoupled localized spatial correlations from sequential connection state transitions. Conv1D filters (64 filters, kernel size 3) extracted cross-packet feature dependencies, while LSTM cells (64 units) captured multi-second sequence dynamics. On the held-out KDDTest+ benchmark, the model achieved 96.89% precision on benign operational telemetry and 89.04% recall on volumetric DoS floods (Macro F1: 0.7127, AUC-ROC: 0.8471).

Answer to RQ3 (Edge Real-Time Execution and Quantization): Post-training Float16 quantization successfully compressed the neural network binary from 4.88 MB to 0.82 MB (an 83.2% reduction). On a physical 1 GB RAM Raspberry Pi 4B edge gateway, single-sample inference latency dropped from 157.66 ms (baseline) to 0.76 ms (a 207-fold speedup). This executes 131 times faster than the 100 ms SCADA deadline and comfortably satisfies the 20 to 50 ms cyclic scan loop deadlines of mining PLCs at a minimal power draw of 2.5 Watts.

Answer to RQ4 (Empirical Transferability and Economic Impact): Transfer learning evaluation on the physical 51-sensor SWaT SCADA testbed demonstrated 59.95% accuracy and an AUC-ROC of 0.8650 in 0.12 ms without retraining, confirming cross-process transferability. Economic risk modeling confirmed that deploying this open-source framework across crushing, milling, and ventilation circuits delivers an estimated return on investment exceeding 200x, mitigating unplanned downtime losses of \$300,000 to \$450,000 per incident while eliminating life-safety risks.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a metaheuristic-optimized, edge-deployable deep learning framework for intrusion detection in mining IoT and SCADA networks. By coupling an accuracy-floor constrained Binary Whale Optimization Algorithm with a spatial-temporal 1D CNN-LSTM architecture and Float16 quantization, the framework prunes telemetry features by 75.61% and achieves a 0.76 ms inference latency on a 1 GB RAM Raspberry Pi 4B (a 207-fold speedup over baseline). The system maintains 96.89% precision on benign telemetry and 89.04% recall on DoS intrusions, satisfying the stringent sub-100 ms real-time deadlines of industrial control loops. Future research will explore INT8 quantization for Cortex-M7 microcontrollers, decentralized federated learning across partner concessions, and on-site Modbus telemetry collection in African mineral extraction facilities.

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